Data Collection and Missing Data

Announcements

- Chapter 8 and 18

Notation

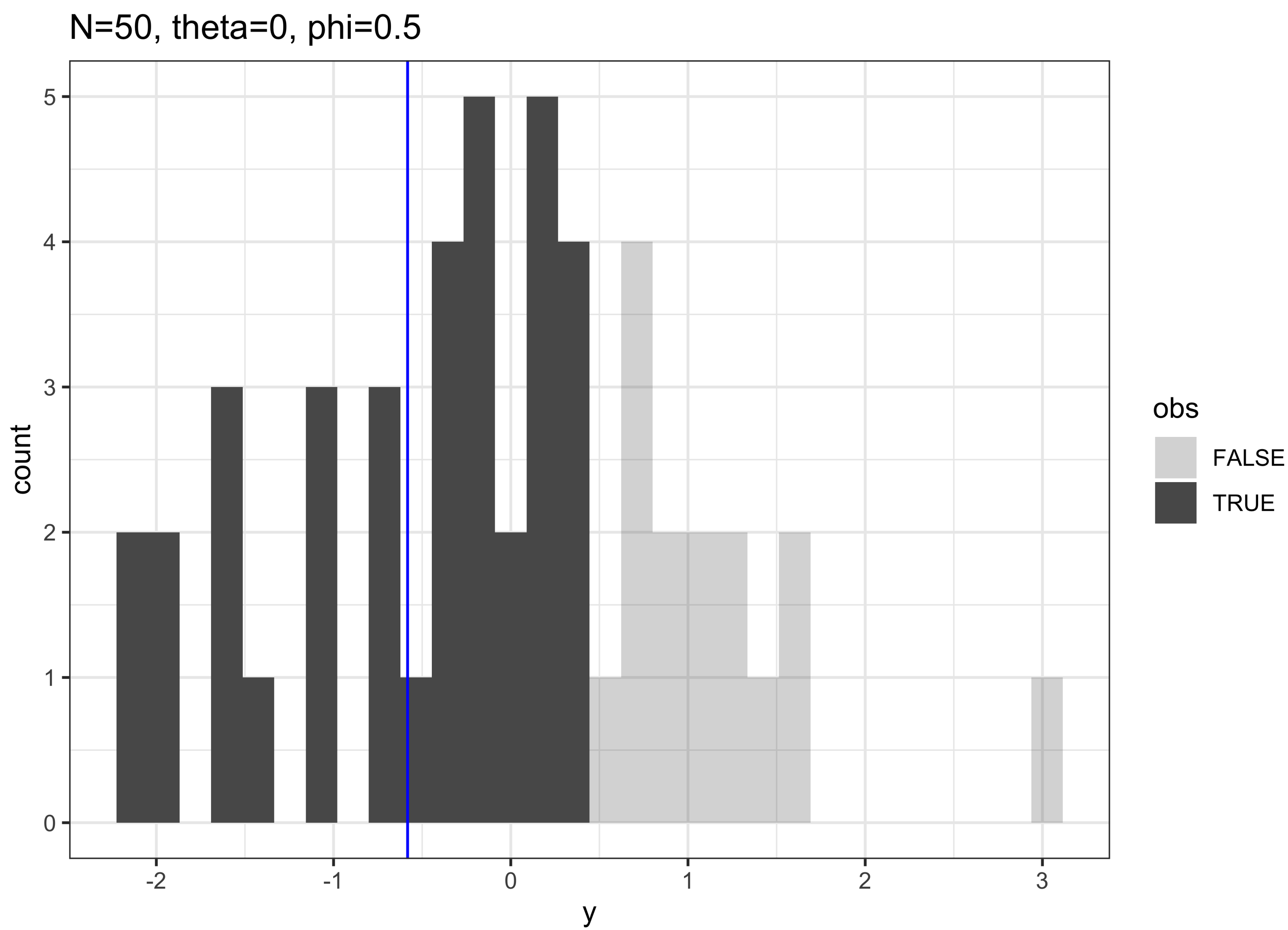
- $y = (y_1, \ldots, y_N)$, y is a matrix (y_i can be a p-vector.
- $I = (I_1, \ldots, I_N)$ be a matrix of the same dimensions as y
- ‘obs’= $\{(i, j) : I_{ij} = 1\}$ and ‘mis’= $\{(i, j) : I_ij = 0\}$

Ignorability

Ignorability

- Two conditions imply ignorability:
  - Missing at random
  - Distinct parameters

Censoring and Truncation



Censoring (known point)

Censoring (known point)

```
data {
  real ybar;
  int<lower=0> truncation_point;
  int<lower=0> ntotal;
  int<lower=0> nobs;
} parameters {
  real theta;
}
model {
  target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) +
  (ntotal-nobs)*log(Phi(theta - truncation_point));

}
```

```
Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.
Chain 2 finished in 0.0 seconds.
Chain 3 finished in 0.0 seconds.
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.2 seconds.

# A tibble: 2 × 10
  variable    mean  median    sd   mad      q5      q95  rhat ess_bulk ess_tail
  <chr>      <dbl>   <dbl> <dbl> <dbl>   <dbl>    <dbl> <dbl>    <dbl>    <dbl>
1 lp__     -15.2   -14.9   0.703 0.319 -16.6   -14.7     1.00    1955.    2380.
2 theta     -0.184  -0.179 0.150 0.151  -0.430   0.0606  1.00    1470.    2229.
```

Censoring (unknown point)

Censoring (unknown point)

```
data {
  real ybar;
  real<lower=ybar> ymax;
  real<lower=0> ntotal;
  real<lower=0> nobs;
} parameters {
  real theta;
  real<lower=ymax> phi;
}
model {
  target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) +
  (ntotal-nobs)*log(Phi(theta - phi));
}
```

```
Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.
Chain 2 finished in 0.0 seconds.
Chain 3 finished in 0.0 seconds.
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.1 seconds.
```

Unknown censoring point

```
# A tibble: 3 × 10
  variable     mean   median     sd    mad      q5     q95  rhat ess_b…¹ ess_t…²
  <chr>       <dbl>    <dbl>  <dbl>  <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl>
1 lp__     -26.0    -25.7    1.07   0.785  -28.2   -25.0     1.00   1591.   1884.
2 theta     -0.0742  -0.0777 0.145  0.148   -0.310   0.172  1.00   2426.   2493.
3 phi        0.468    0.451  0.0548 0.0408   0.415   0.579  1.00   1708.   1292.
# … with abbreviated variable names ¹ess_bulk, ²ess_tail
```

Truncation (known point)

```
data {
  real ybar;
  real<lower=ybar> ymax;
  real<lower=0> ntotal;
  real<lower=0> nobs;
  real<lower=ymax> truncation_point;
} parameters {
  real theta;
}
model {
  target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) -
  nobs*log(1.0 - Phi(theta - truncation_point));
}
```

Truncation (known point)

```
Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.
Chain 2 finished in 0.0 seconds.
Chain 3 finished in 0.0 seconds.
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.2 seconds.
```

```
# A tibble: 2 × 10
  variable   mean median    sd   mad     q5   q95  rhat ess_bulk ess_tail
  <chr>     <dbl>  <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>    <dbl>    <dbl>
1 lp__      7.56   7.81  0.652 0.293  6.27  8.02   1.00    1775.    2194.
2 theta    -0.123 -0.123 0.225 0.228 -0.493 0.245  1.00    1485.    1625.
```

Truncation (unknown point)

Truncation (unknown point)

```
data {
  real ybar;
  real<lower=ybar> ymax;
  real<lower=0> ntotal;
  real<lower=0> nobs;
} parameters {
  real theta;
  real<lower=ymax> truncation_point;
}
model {
  truncation_point ~ normal(0, 10);
  target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) -
  nobs*log(1.0 - Phi(theta - truncation_point));
}
```

Truncation (unknown point)

```
Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.
Chain 2 finished in 0.0 seconds.
Chain 3 finished in 0.0 seconds.
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.1 seconds.
```

```
# A tibble: 3 × 10
  variable           mean median    sd    mad     q5   q95  rhat ess_b…¹ ess_t…²
  <chr>             <dbl>  <dbl> <dbl>  <dbl>  <dbl> <dbl> <dbl>   <dbl>   <dbl>
1 lp__              4.53   4.89  1.11  0.802   2.19  5.60   1.00   1239.   1005.
2 theta            -0.124 -0.126 0.258 0.255  -0.536 0.299  1.00   1014.    879.
3 truncation_point  0.667  0.460 1.50  0.0493  0.416 0.743  1.00   1170.    750.
# … with abbreviated variable names ¹ess_bulk, ²ess_tail
```

Multivariate Normal

MVN with missingness in STAN

```
Running MCMC with 4 parallel chains...

Chain 4 finished in 3.2 seconds.
Chain 2 finished in 3.4 seconds.
Chain 3 finished in 3.6 seconds.
Chain 1 finished in 3.8 seconds.

All 4 chains finished successfully.
Mean chain execution time: 3.5 seconds.
Total execution time: 3.9 seconds.
```